

AN ECLECTIC TECHNOLOGIES FOUNDATIONAL DEFINITION

FD-LD — Layer Decomposition and Determinism

Foundational Definition: The Two-Layer Architecture, the Determinism Boundary as a Scope Gate, and the Machine-Learning Carve-Out

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Designation: FD-LD **Family:** Foundational Definitions **Version:** 0.1 — DRAFT (promoted from, and superseding, the *Layer Decomposition and Determinism Terminology Annex v0.1*) **Date:** August 2026 **Author:** Timothy E. Roxey **Organization:** Eclectic Technologies **Audience:** Asset owner/operators; ICS vendors and suppliers; regulatory staff and standards-community reviewers **Parent:** None. This instrument defines. See §2. **Companion root definition:** FD-BL, which owns the placement criterion. FD-BL §4.4 names this instrument as the source of its scope gate. FD-LD supplies the determinism and layer vocabulary; it does **not** own placement.

1. Purpose and the Defect This Instrument Corrects

The annex this instrument replaces did one thing right and one thing wrong, and the two are separable. Right: it unified two vocabularies. The Eclectic Technologies governance work describes one determinism boundary in two languages — the ISA / design-basis vocabulary of infrastructure and cognitive layers, and the determinism vocabulary of deterministic and non-deterministic systems. Stating that these are the same boundary, so a reader arriving from either recognizes the other, is correct and useful. That unification is retained here in full (§3).

Wrong: it fused that boundary with the placement criterion. The annex’s mapping table carried a third row equating “the boundary between the layers” with “the Bright Line placement criterion,” resolving both to “whether a system’s behaviour can be bounded by proof,” and asserting that “everything downstream — Command Broker interposition, assessment method, tier rigour — follows from which side a component falls on.” That row makes determinism the criterion for *where the line falls*. FD-BL §4.2 and §4.4 hold the opposite: determinism decides **scope** — which components are subject to Bright Line analysis — and **consequence** decides **placement** — where the line falls for the actions a component produces.

The consequence of the fusion was not local. Because the annex is the cited authority for the sector series (issued under the ISA Parent Framework consistency protocol), the fused row propagated as doctrine: DBA-DC-L1 §3.5, DBA-EN-SF, and DBA-ES all state placement on the determinism boundary, and the Command Broker Framework §3.2 carries both the fused doctrine and, contradicting it, the correct consequence criterion. This instrument withdraws the fused row, keeps everything else, and thereby removes the doctrine at its source.

2. Authority, Family, and Convention

2.1 Precedence

FD-LD owns, for the whole corpus, the two-layer decomposition, the determinism vocabulary, and the machine-learning carve-out — that is, the **scope gate** that decides which components are subject to Bright Line analysis and the Command Broker obligation.

FD-LD does **not** own placement. Where the Bright Line falls is owned by FD-BL (§4.2) for the industrial corpus and by FD-SBL for the surveillance corpus. Any instrument that states, cites, or restates the determinism or layer boundary as the *placement criterion* varies from this instrument and shall be conformed. This instrument supersedes the Layer Decomposition and Determinism Terminology Annex v0.1; references to that annex resolve here.

2.2 Family membership

FD-LD qualifies as a Foundational Definition: the correctness of the sector frameworks, the facility-class instruments, and the Command Broker Framework depends on agreeing what kind of thing the determinism boundary is — a scope gate, not a placement line.

2.3 Declaration convention

Per the family convention, the ontology is declared first (§4.1): the determinism boundary is a **scope boundary**. It partitions components by whether their behaviour can be bounded by proof, in order to decide which components are subject to Bright Line analysis. It is not itself the Bright Line and not the criterion for where the Bright Line falls.

2.4 What this instrument does not alter

The two-layer decomposition, the determinism vocabulary, the machine-learning carve-out, and the treatment of deterministic protective functions are inherited from the annex unchanged in substance (§3, §4.3). Only the equation of the layer boundary with placement (the annex's third table row) is withdrawn.

3. The Two Vocabularyes, One Scope Boundary

The following is retained from the annex, corrected only in its header row and in the removal of the placement equation.

ISA / design-basis

vocabulary	Determinism vocabulary	The single underlying property
Infrastructure layer	Deterministic systems; deterministic machine learning within bounded inputs	Behaviour is reproducible and formally verifiable. The output space can be bounded and exhaustively reasoned about. Existing standards — ISA/IEC 62443, IEC 61508, IEEE — are adequate to it.
Cognitive layer	Generative AI; non-deterministic systems	The same input may produce different outputs across execution cycles. The response space cannot be exhaustively enumerated. Formal verification does not apply; empirical assessment is required.

The infrastructure/cognitive distinction and the deterministic/non-deterministic distinction are the same boundary in two vocabularyes. That boundary is a **scope boundary**: which side a component falls on

decides whether it is subject to Bright Line analysis and the Command Broker obligation. The annex's third row — which added “The Bright Line placement criterion” to this equivalence — is withdrawn (§4.2).

4. The Definition

4.1 What kind of thing the determinism boundary is

The determinism boundary is a scope gate. It partitions components by whether their behaviour can be bounded by proof, and it decides which components are subject to Bright Line analysis. A component whose behaviour is deterministic within a stated input bound and formally testable as deployed is on the infrastructure side and outside the analysis; a component whose behaviour cannot be bounded by proof is on the cognitive side and within it.

The boundary is not the Bright Line, and which side a component falls on does not decide where the line falls for that component's actions. It decides only whether those actions are analysed under the line at all.

4.2 The correction: the layer boundary is not the placement criterion

The annex asserted that the layer boundary is the Bright Line placement criterion, and that Command Broker interposition follows from which side a component falls on. That assertion is withdrawn.

Determinism decides scope. Consequence decides placement (FD-BL §4.2). Fusing them produces two errors at once, both named in FD-BL §4.4: it implies that every action of a non-deterministic component is above the line — including actions of no material consequence — and it implies that a deterministic component can never produce an above-line action, which is false and which removes from analysis the automated systems most capable of causing harm. A deterministic component that alters a protective function's setpoints takes an above-line action on the consequence of that alteration; its determinism governs only whether *its own* behaviour is subject to the obligation, not the placement of an action taken upon it (FD-BL §4.4).

Command Broker interposition therefore does not follow from “which side a component falls on.” It follows from two separate determinations in sequence: whether the component is in scope (this instrument), and whether a given action is above the line on its consequence (FD-BL §4.2, action by action per §4.3).

4.3 The machine-learning carve-out

A component is exempt from cognitive-layer assessment only if it is deterministic within a **stated input bound** and formally testable in the deployed configuration. The qualifier is essential: an exemption asserted as “merely deterministic” without tying the claim to the range over which determinism was established is not a valid carve-out.

The test binds in both directions. A component is not exempted by being described as machine learning; a model whose outputs are not reproducible under identical inputs is a cognitive-layer system whatever its architecture is named. Any component relying on the carve-out shall have its input bound stated and verified as part of assessment.

Deterministic protective functions — reactor trip, engineered safety-feature actuation, protective-relay operation — are already automated and deterministic. They are outside the Command Broker obligation, but **by the carve-out (scope), not by being below the line**. Asserting they are below the line would be false, because they are among the highest-consequence actions in any facility; they are

outside the obligation because their behaviour is bounded by proof (FD-BL §4.5). This distinction is the whole point of separating scope from placement, and it is why the fused row had to go.

4.4 Scope, placement, and obligation are three questions

FD-LD owns the first. Which components are analysed at all is scope (this instrument). Where the line falls for an analysed component's actions is placement, on consequence (FD-BL §4.2). Who authorises an above-line action is the obligation, discharged by the Broker (FD-BR). No one of the three answers another.

Cross-corpus note. FD-LD supplies the determinism and layer vocabulary to both corpora. The carve-out — deterministic components are out of scope — applies to the industrial Bright Line. FD-SBL §4.4 explains why the carve-out does not transfer to the surveillance Bright Line: a deterministic statistical model's societal effect is not bounded by proof, so determinism is not a valid scope gate there. FD-LD's vocabulary is shared; its carve-out is industrial.

5. Worked Examples

A supervised anomaly-detection model on a feedwater pump, deterministic within a stated input bound and formally testable as deployed, is an infrastructure-layer component: outside Bright Line analysis (§4.3). Replace it with a generative system producing the same recommendations, and it becomes a cognitive-layer component: within scope, its actions then placed on consequence (FD-BL §4.2).

A vendor asserts a component is “deterministic” but cannot state the input range over which that holds. The assertion fails the qualifier (§4.3); the component is treated as cognitive-layer until the bound is stated and verified.

A reactor-trip function performs an action of the highest consequence without a Command Broker. It is not below the line; it is outside the obligation by the carve-out, because its behaviour is bounded by proof (§4.3). This is the example that makes the scope/placement distinction unavoidable.

6. Conformance

The instruments that inherited the fused doctrine from the superseded annex conform to FD-LD by separating scope from placement:

DBA-DC-L1 §3.5 (series root), **DBA-EN-SF**, **DBA-ES** — restate the determinism boundary as the scope gate and place the line on consequence per FD-BL §4.2. Detailed diagnosis and replacement text are in FDBL-CS-P2.

Command Broker Framework §3.2 — resolve the layer/consequence contradiction in favour of consequence; the determinism boundary is scope. Markup drafted.

ISA Parent Framework — the annex was issued under its consistency protocol; update the reference to cite FD-LD and confirm the ISA Parent's own placement language (Pass 2 of the coherence scan). The annex itself is superseded and withdrawn.

7. Author Decisions Outstanding

Confirmation of the designation **FD-LD** and its filing with the Foundational Definitions family, which FD-BL §4.4 already anticipates.

Whether the “two vocabularies, one boundary” unification (§3) is retained as drafted — scoped to the scope boundary — or restructured. It is kept here because it is correct and useful once placement is removed from it.

The ISA Parent Framework consistency-protocol clause under which the annex was issued should be re-pointed at FD-LD, and the ISA Parent’s own statement of the boundary confirmed against this instrument.

8. References

FD-BL — *The Bright Line*. Version 0.3. Eclectic Technologies, 2026. (Owns placement; names FD-LD as its §4.4 scope source.)

FD-SBL — *The Surveillance Bright Line*. Version 0.2. Eclectic Technologies, 2026. (§4.4 explains why the determinism carve-out does not transfer to the surveillance domain.)

FD-BR — *The Broker*. Version 0.1. Eclectic Technologies, 2026.

Layer Decomposition and Determinism Terminology Annex. Version 0.1. Eclectic Technologies, 2026. (Superseded by this instrument.)

ISA/IEC 62443; IEC 61508; IEEE standards for deterministic industrial systems (§3, infrastructure layer).

Revision History

0.1 — August 2026. Promoted from the *Layer Decomposition and Determinism Terminology Annex* v0.1 to a Foundational Definition. Retained the two-layer decomposition (§3), the determinism vocabulary, the machine-learning carve-out (§4.3), and the deterministic-protective-function treatment. Withdrew the annex’s third mapping row equating the layer boundary with the Bright Line placement criterion, and declared the determinism boundary a scope gate (§4.1) distinct from placement (§4.2), consistent with FD-BL §4.2/§4.4. Added the cross-corpus note (§4.4) that the carve-out is industrial and does not transfer to FD-SBL. This instrument removes, at its source, the conflation identified in the corpus coherence scan (FDBL-CS-SCAN-1) as the doctrinal origin of the placement error.